⭐ What is Virtualization?

Virtualization is a technology that allows one physical computer/server to run multiple virtual machines (VMs).
Each VM works like a separate computer with its own CPU, RAM, storage, and operating system.

🔍 Why do we use Virtualization?

Without virtualization:

- One server runs only one OS or one application → hardware is wasted.

With virtualization:

- One server can run many OS and many applications → better performance, less cost.

🏗 How does Virtualization work?

It uses a special software called a **Hypervisor**.

✔ Hypervisor:

- Sits between hardware and VMs
- Distributes CPU, RAM, storage to each VM
- Makes every VM feel like it has its own hardware

🧩 Types of Hypervisors

⭐ Type 1: Bare Metal Hypervisor

- Runs directly on hardware
- Faster and more secure
  Examples: VMware ESXi, Microsoft Hyper-V, KVM

⭐ Type 2: Hosted Hypervisor

- Runs on top of an operating system
  Examples: VirtualBox, VMware Workstation

🎯 Advantages of Virtualization

✔ Saves hardware cost
✔ Better resource utilization

✔ Easy backup and recovery
✔ Fast to create or remove VMs
✔ Isolation (one VM doesn't affect others)

⭑ What is a VM?

A **VM is a virtual computer created using virtualization software.**
It has its own:

- CPU
- RAM
- Storage+
-
- Operating System (Windows, Linux, etc.)

Even though it uses the hardware of the real physical machine, it behaves like a completely **separate computer**.

🧠 Example

You have a laptop with Windows.
Inside it, you create a VM and install Linux.

Now you have:

- Windows (host OS)
- Linux VM (guest OS)

Both run at the same time on one machine.

🧩 Real-life example in cloud

AWS EC2 = A virtual machine
Azure VM = A virtual machine
Google Compute Engine = A virtual machine

Hypervisor

A **hypervisor** is a **software layer** that allows you to create and run **multiple virtual machines (VMs)** on a single physical machine.

✅ Simple Definition

Hypervisor = Virtualization software that creates and manages VMs.

It separates:

- **Hardware (CPU, RAM, storage)**
  from
- **Virtual Machines (VMs)**

⭐ What is Cloud Computing?

Cloud Computing is the delivery of computing services over the internet.
Instead of buying your own servers or hardware, you use computing resources provided by cloud companies like **AWS, Azure, Google Cloud**.

These resources include:

- Servers (VMs)
- Databases
- Storage
- Networking
- Security
- AI/ML services
- Analytics
- DevOps tools

All delivered **on demand**, you pay only for what you use.

🎯 Real-life Examples

- Netflix uses AWS to stream movies
- Paytm uses cloud to handle payments
- Gmail is cloud-based email
- PUBG uses cloud servers for gaming

🧠 Simple Definition

Cloud Computing = Renting IT resources over the internet instead of owning them.

Example:
Your laptop memory full?
Upload to Google Drive → That's cloud storage.

Your company needs servers?
Use AWS EC2 → That's cloud computing.

⭐ Types of Cloud Services (Service Models)

1️⃣ IaaS – Infrastructure as a Service

Definition:
IaaS provides virtualized computing resources over the internet such as **virtual machines, storage, networks, and firewalls**.

Use case:
When companies want full control over the server but don't want to buy hardware.

Examples:
AWS EC2, Google Compute Engine, Azure Virtual Machines.

2️⃣ PaaS – Platform as a Service

Definition:
PaaS provides a **platform** to build, run, and deploy applications without managing the underlying infrastructure.

You get the runtime environment, OS, database, and web server ready.

Use case:
Developers want to focus only on writing code, not server setup.

Examples:
AWS Elastic Beanstalk, Google App Engine, Azure App Service.

3️⃣ SaaS – Software as a Service

Definition:
SaaS provides **ready-to-use software** over the internet on a subscription basis.
Users don't manage anything—just use the application.

Use case:
Email, CRM, file storage, collaboration tools.

Examples:
Gmail, Google Drive, Salesforce, Microsoft 365.

Types of Cloud Deployment Models

1️⃣ Public Cloud

Definition:
The cloud infrastructure is owned and managed by third-party cloud providers and delivered over the internet.

Features:

- Resources are shared among multiple users (multi-tenant).
- Cost-effective and scalable.
- No maintenance required.

Examples:
AWS, Microsoft Azure, Google Cloud.

Use Case:
Startups, SaaS products, scalable applications.

2️⃣ Private Cloud

Definition:
The cloud infrastructure is used exclusively by a single organization.

Features:

- Higher security and control.
- Hosted on-premise or in a dedicated data center.
- Suitable for sensitive workloads.

Examples:
VMware Private Cloud, OpenStack.

Use Case:
Banks, government organizations, enterprises with strict compliance.

3️⃣ Hybrid Cloud

Definition:
A combination of public and private clouds that are connected to share data and applications.

Features:

- Flexibility to keep sensitive data in private cloud.
- Ability to scale to public cloud when needed (cloud bursting).
- Best of both worlds.

Examples:
AWS Outposts, Azure Hybrid Cloud.

Use Case:
Large enterprises, disaster recovery, variable workloads.

⭑ What is AWS?

AWS (Amazon Web Services) is the world's largest **cloud computing platform** that provides on-demand services like computing power, storage, databases, networking, security, AI/ML, and DevOps tools over the internet.

With AWS, companies don't need to buy physical servers—they can **rent cloud services** and pay only for what they use.

"AWS is a secure, scalable, and cost-effective cloud platform by Amazon that offers more than 200 services such as EC2 for compute, S3 for storage, RDS for databases, and Lambda for serverless computing."

⭐ Why do we use AWS?

✔ No hardware to buy or maintain
✔ Highly scalable (auto scaling)
✔ Pay-as-you-go
✔ Global availability with data centers
✔ High security
✔ Fast deployment

🔧 Popular AWS Services

- **EC2** → Virtual servers
- **S3** → Object storage
- **RDS** → Managed databases
- **Lambda** → Serverless compute
- **VPC** → Networking
- **CloudFront** → CDN
- **DynamoDB** → NoSQL database

⭐ What Makes AWS Special?

AWS stands out from other cloud providers because of its **maturity, global scale, huge service variety, reliability, and strong security**. It was also the **first major cloud platform**, so it has the largest ecosystem and customer base.

⭐ AWS in the Present Market (Interview Answer)

AWS is currently the **market leader** in cloud computing. It holds the **largest market share** compared to competitors like Microsoft Azure and Google Cloud. AWS dominates the cloud industry because of its **mature platform, wide service offerings, global infrastructure, and strong enterprise adoption**.

1. Market Leadership

AWS is the **first and largest cloud provider**.
Most Fortune 500 companies use AWS for hosting, storage, analytics, and AI.

Even today, AWS remains the **top choice** for:

- Startups
- Enterprises
- Government organizations
- Global tech companies

🌍 AWS Global Infrastructure — Interview Answer

“AWS global infrastructure consists of Regions, Availability Zones, Edge Locations, Local Zones, and Wavelength Zones. This distributed design gives AWS high availability, low latency, global reach, and strong fault tolerance—making it the most reliable cloud platform in the world.”

Availability Zones (AZs)

An **Availability Zone** is a group of **one or more isolated data centers** inside a region

Example:
Mumbai region (ap-south-1) has 3 AZs: ap-south-1a, 1b, 1c.

Local Zones

Local Zones bring AWS services **closer to major cities**, reducing latency for applications like gaming, real-time video, or ML.

Example:
AWS Local Zone in Delhi.

🔐 What is AWS IAM? (Interview Answer)

AWS IAM (Identity and Access Management) is a service that helps you **securely control access** to AWS resources.

With IAM, you can decide:

- **Who** can access AWS (Users, Groups, Roles)
- **What** they can access (S3, EC2, RDS, etc.)
- **How much** access they have (Read, Write, Full)

IAM is completely **free** and is one of the most important security services in AWS.

⭐ Key Features of IAM

1️⃣ IAM Users

Individual people or applications that need AWS access.

2️⃣ IAM Groups

A collection of users.
You assign permissions to the group, not each user.

3️⃣ IAM Roles

Temporary access for:

- EC2 instances
- Lambda functions
- Third-party apps
- Cross-account access

No password required—only permissions.

4️⃣ IAM Policies

JSON documents that define permissions.
Example: Allow S3 read-only access.

🛡 Security Features

- MFA (Multi-Factor Authentication)
- Password policies
- Access key rotation
- Least privilege principle
- CloudTrail logging for monitoring

🔐 What is MFA (Multi-Factor Authentication)?

MFA is an extra layer of security that requires users to provide **two or more** types of verification to log in.

Normally, you use only a **password**, but with MFA you also need a **second factor**, like:

- A code from a mobile authenticator app
- A text message OTP
- A hardware security key

This makes your account much more secure.

⭑ Ways of Accessing AWS

1️⃣ AWS Management Console

- A **web-based UI** to manage AWS services.
- Best for beginners and admin tasks.
- You login with username/password + MFA.

Example:
Creating EC2, S3 bucket through dashboard.

2️⃣ AWS CLI (Command Line Interface)

- A **text-based command tool** for AWS.
- Used by developers, DevOps engineers, and automation scripts.

- You configure it using **Access Key + Secret Key**.

Example:
`aws s3 ls` — to list S3 buckets.

3️⃣ AWS SDKs (Software Development Kits)

- Used to interact with AWS **programmatically** inside applications.

⭐ What is AWS EC2?

AWS EC2 (Elastic Compute Cloud) is a cloud service that provides **virtual servers** on demand.
It allows you to run applications by launching virtual machines called **EC2 Instances**.

Think of EC2 as:
☞ *"Renting a computer/server from AWS that you can use anytime, and pay only for the hours you use."*

🔍 Why is it called Elastic?

Because you can:

- **Scale up** (more power)
- **Scale down** (less power)
- **Add or remove instances** automatically

based on demand.

⭐ Key Features of AWS EC2

1️⃣ Different Instance Types

Choose CPU, RAM, storage based on need.
Example:

- t2.micro → small workloads
- m5.large → general apps

2️⃣ Multiple OS Support

You can run:

- Windows
- Linux

4️⃣ Storage Options

- EBS (Elastic Block Storage)

- Instance Store

🔐 What is a Security Group in EC2?

A **Security Group** is a **virtual firewall** for your EC2 instance that **controls inbound and outbound traffic**.

It decides:(By default launch wizard 4)

- **Which traffic is allowed to enter** your EC2 instance (Inbound rules)
- **Which traffic is allowed to go out** from your EC2 instance (Outbound rules)

Security Groups work at the **instance level**, not the VPC or subnet level.

Key Points :

✔ 1. Acts as a Virtual Firewall

Security Groups protect EC2 instances by allowing or denying traffic based on rules

✔ 2. Inbound & Outbound Rules

- **Inbound:** Defines what traffic can come *into* the instance
  Example: Allow port **22** for SSH, port **80** for HTTP, port **443** for HTTPS
- **Outbound:** Defines what traffic can go *out* of the instance
  (By default, all outbound traffic is allowed)

✔ 3. Default Behavior

- **All inbound traffic is blocked**
- **All outbound traffic is allowed**

✔ 4. Stateful

If you allow inbound traffic on a port, the response is automatically allowed, even if outbound rule doesn't specify it.

6. Only "Allow" Rules

Security Groups **do NOT** support "deny" rules.
They only allow traffic.

How to Connect to an EC2 Instance ?

If You're Using Windows

Use PuTTY

- Convert `.pem` → `.ppk` using PuTTYgen
- Connect with username (`ec2-user`, `ubuntu`, etc.)

Are EC2 Instances Region Specific?

An EC2 instance is always created in a specific AWS Region **.**

Yes, EC2 instances are region-specific. You must select the same region where the instance was created to view or connect to it.

EC2 Instance Types and Their Use Cases

1️⃣ General Purpose Instances

Examples: `t2`, `t3`, `t4g`, `m5`, `m6g`
Use Case: Balanced for CPU + RAM

2️⃣ Compute Optimized Instances

Examples: `c4`, `c5`, `c6g`, `c7g`
Use Case: CPU-heavy workloads

- High-performance computing (HPC)
- Gaming servers

3️⃣ Memory Optimized Instances

Examples: `r5`, `r6g`, `x1e`, `u-6tb1`
Use Case: RAM-heavy workloads

- Large databases (MySQL, Postgr

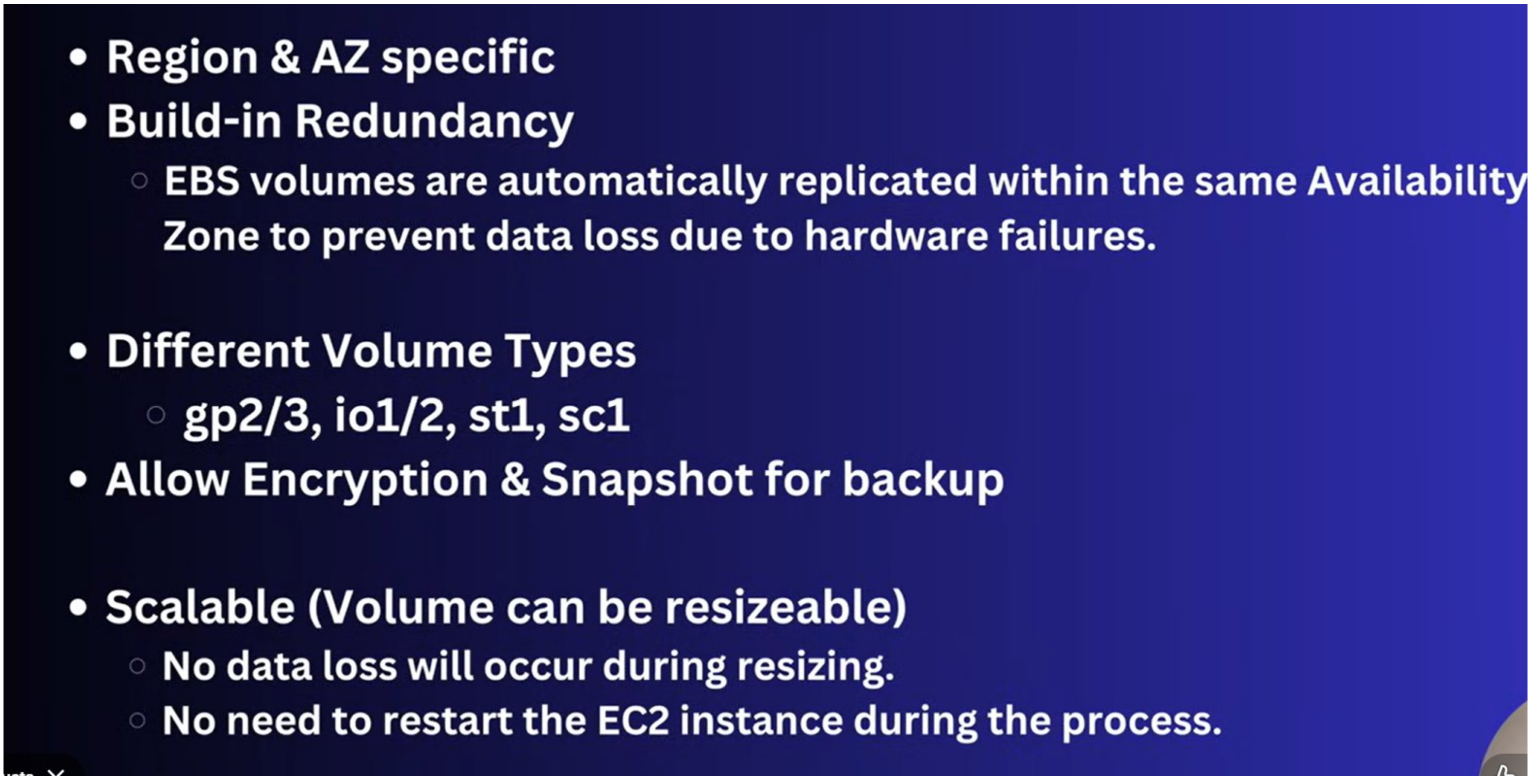
AWS EBS (Elastic Block Store)

✅ What is EBS?

EBS is **block-level storage** for EC2 instances.
It works like a **virtual hard disk** (SSD/HDD) attached to your EC2 instance.

- Stores OS, applications, logs, databases
- Data does NOT delete when the instance stops
- Highly durable and reliable

✔ Key Features of EBS



Practical : connecting EBS with Ec2 instances.

INTERVIEW QUESTIONS

What is Cloud Computing?

Cloud computing is the on-demand delivery of computing resources—like servers, storage, databases, networking, and software—over the internet on a pay-as-you-go model. It removes the need to manage physical hardware and lets users scale resources instantly based on demand

✅ What is AWS?

AWS (Amazon Web Services) is the world's largest cloud computing platform that provides on-demand services like compute power, storage, databases, networking, security, and analytics on a pay-as-you-go basis.

It allows companies to build and deploy applications without managing physical servers.

✅ Examples of AWS Services (Very Common Question)

- **EC2** – Virtual servers
- **S3** – Object storage
- **RDS** – Managed database service
- **VPC** – Virtual private cloud/network

5 advantages of using aws over on premise server ?

✅ 1. No Upfront Hardware Cost

✅ 2. Scalability & Elasticity.

✅ 3. High Availability & Reliability

✅ 4. Faster Deployment

✅ 5. Global Reach

"What are the main cloud service models?"

✅ Main Cloud Service Models (Interview Answer)

1. **IaaS (Infrastructure as a Service)**
   Provides virtualized computing resources like servers, storage, and networks.
   Example: AWS EC2, S3
2. **PaaS (Platform as a Service)**
   Provides a platform to develop, test, and deploy applications without managing underlying infrastructure.
   Example: AWS Elastic Beanstalk, Google App Engine
3. **SaaS (Software as a Service)**
   Provides fully managed software applications accessible through a browser.
   Example: Gmail, Salesforce, Microsoft 365

Differences between IaaS, PaaS, and SaaS

Feature	IaaS	PaaS	SaaS
Full Form	Infrastructure as a Service	Platform as a Service	Software as a Service
What It Provides	Virtual servers, storage, networking	Platform for developing & deploying apps	Ready-to-use software
User Controls	OS, storage, applications	Applications & data only	Only end-user settings
Who Manages Infra?	Provider manages hardware	Provider manages hardware + OS	Provider manages everything
Use Case	Hosting servers, VMs, networks	App development without managing servers	Email, CRM, office tools
Examples	AWS EC2, Azure VM	AWS Elastic Beanstalk, Heroku	Gmail, Salesforce

"What is the AWS Shared Responsibility Model?"

The AWS Shared Responsibility Model defines which security tasks are handled by AWS and which are handled by the customer. AWS is responsible for the security of the cloud, while customers are responsible for security in the cloud.

✅ Main Categories of AWS Services (Interview Answer)

AWS services are mainly grouped into the following categories:

1. **Compute**
2. **Storage**
3. **Database**
4. **Networking & Content Delivery**

5. **Security, Identity & Compliance**
6. **Management & Monitoring**
7. **Analytics**
8. **Machine Learning & AI**
9. **Developer Tools**
10. **Serverless**
11. **Migration & Transfer**
12. **Cost Management**

✅ Top 25 AWS Services (Interview-Focused List)

1. EC2 – Virtual servers

2. S3 – Object storage

3. RDS – Managed relational databases

4. VPC – Virtual Private Cloud

5. Lambda – Serverless compute

6. IAM – Identity & Access Management

7. CloudWatch – Monitoring & logging

8. CloudTrail – API auditing

9. ELB (Load Balancer) – Traffic distribution

10. Auto Scaling – Scale resources automatically

11. DynamoDB – NoSQL database

12. Route 53 – DNS & domain management

13. EBS – Block storage for EC2

14. SNS – Notifications service

15. SQS – Message queue service

16. ECR – Container registry

17. ECS – Container orchestration

18. EKS – Managed Kubernetes

19. CloudFormation – IaC (Infrastructure as Code)

20. Elastic Beanstalk – App deployment platform

21. KMS – Encryption key management

22. GuardDuty – Threat detection

23. Redshift – Data warehousing

24. Glue – ETL & Data integration

25. API Gateway – Manage and deploy APIs

✅ Hybrid Cloud (Interview Answer)

A hybrid cloud is a combination of on-premise infrastructure and public cloud services that are integrated to work together. It allows data and applications to move between the on-premise environment and the cloud seamlessly.

✅ When to Use Hybrid Cloud (Interview Answer)

You should use a **hybrid cloud** when:

1. **You need to keep sensitive data on-premise**
   (e.g., banking, healthcare, government compliance)
2. **You want to use cloud for scaling but keep core systems in-house**
   Example: Run daily workloads on-prem but burst to cloud during peak load.
3. **You have legacy applications that cannot be fully moved to cloud**
   Hybrid cloud supports a partial migration strategy.
4. **You want business continuity and disaster recovery**
   Keep backup in cloud while primary workload runs on-prem.
5. **You need cost optimization**

✅ What is a Resource in AWS? (Interview Answer)

A resource in AWS is an individual component or entity that create or use inside a service.
Example:

- An **EC2 instance**
- An **S3 bucket**
- A **Lambda function**
- A **VPC subnet**

These are **actual objects** that you provision and manage.

✅ What is a Service in AWS? (Interview Answer)

A service in AWS is a category or product offered by AWS that provides specific functionality.
Example:

- **EC2** (Compute service)
- **S3** (Storage service)
- **RDS** (Database service)

Services provide the **capability**, and resources are the **actual instances** created using that capability

✅ What is an AWS Region? (Interview Answer)

An AWS Region is a separate geographic area where AWS has multiple data centers. Each region is isolated for fault tolerance and compliance needs.
Example:

- **us-east-1 (N. Virginia)**
- **ap-south-1 (Mumbai)**

✅ What is an Availability Zone (AZ)? (Interview Answer)

An Availability Zone is one or more physically separate data centers *within a region*, each having its own power, cooling, and networking.
AZs are connected with low-latency, high-speed networks.

Example:

- **ap-south-1a**, **ap-south-1b**, **ap-south-1c**

✅ Top 5 AWS Compute Services (Most Asked in Interviews)

1️⃣ Amazon EC2 (Elastic Compute Cloud)

Virtual servers used to run applications.

2️⃣ AWS Lambda

Serverless compute service that runs code without provisioning servers.

3️⃣ Amazon ECS (Elastic Container Service)

Container orchestration for running Docker containers.

4️⃣ Amazon EKS (Elastic Kubernetes Service)

Managed Kubernetes service for deploying containerized applications.

5️⃣ AWS Fargate

Serverless compute engine for containers (works with ECS/EKS).

✅ What are EC2 Instances? (Interview Answer)

EC2 instances are virtual servers in the AWS cloud that we can use to run applications. They provide resizable compute capacity and act like a normal machine where we can install OS, software, and run workloads.

They support different OS like Linux, Windows, Ubuntu, etc.

✅ When to Use EC2 in a Project? (Interview Answer)

You should use **EC2 instances** when:

1. **You need full control over the server**
   (OS configuration, firewalls, custom software installation)
2. **You are running long-running applications**
   (web servers, backend APIs, microservices)

3. **You need to host databases manually**
   e.g., installing MySQL, MongoDB on your own VM
4. **You need predictable and consistent compute power**

✅ Horizontal Scaling (Scale Out)

Adding more servers/instances to handle increased load.

💡 Examples

- Adding more EC2 instances behind a load balancer.

👉 When to Use Horizontal Scaling

Use it when:

- You need **high availability**
- You expect **heavy or unpredictable traffic**

✅ Vertical Scaling (Scale Up)

Upgrading the power of an existing server (CPU, RAM, storage).

💡 Examples

- Moving from **t2.micro → t3.large**
- Increasing RAM/CPU of a database server

👉 When to Use Vertical Scaling

Use it when:

- Your application **cannot run on multiple servers**
- You need quick scaling without architecture changes

✅ What are AWS Auto Scaling Groups / EC2 Auto Scaling?

An Auto Scaling Group (ASG) in AWS automatically adds or removes EC2 instances based on demand to maintain performance, availability, and cost efficiency.

✅ When to Use Auto Scaling (Interview Answer)

Use ASG when you want:

- To handle **unpredictable traffic**
- To ensure **zero downtime**

✅ EC2 Instance Types (Interview Answer)

EC2 instance types are different categories of virtual servers optimized for various workloads, such as general-purpose, compute-heavy, memory-heavy, storage-heavy, and GPU-based applications.

AWS groups EC2 instances based on their purpose and performance characteristics.

✅ Main EC2 Instance Type

Here are the **most commonly asked types**:

1️⃣ General Purpose Instances

Balanced compute, memory & networking.
Examples: t2, t3, t4g, m5, m6g

2️⃣ Compute Optimized Instances

For CPU-heavy workloads.
Examples: c5, c6g

3️⃣ Memory Optimized Instances

For RAM-intensive applications.
Examples: r5, r6g, x1e, z1d

4️⃣ Storage Optimized Instances

For high IOPS, high-throughput storage.
Examples: i3, i4, d2, h1

5️⃣ GPU / Accelerated Computing Instances

For ML, AI, graphics, HPC.
Examples: p3, p4, g4dn, g5

✅ What are On-Demand, Reserved, and Spot Instances in AWS?

✅ 1️⃣ On-Demand Instances (Interview Answer)

On-Demand instances let you pay for EC2 compute with no long-term commitment.
it can launch or terminate any time.

✔ Best When

- Short-term or unpredictable workloads
- Testing or development
- You want flexibility

✅ 2️⃣ Reserved Instances (Interview Answer)

Reserved Instances offer up to 75% discount in exchange for a 1-year or 3-year commitment.
Capacity is reserved in that period.

✔ Best When

- Long-running, steady workloads
- Production servers
- You want predictable cost savings

✅ 3️⃣ Spot Instances (Interview Answer)

Spot Instances let you use spare EC2 capacity at up to 90% discount, but AWS can terminate them anytime with a 2-minute notice.

✔ Best When

- Fault-tolerant, flexible workloads
- Big data, ML training, batch jobs
- Containerized workloads

✅ What is AWS Elastic Beanstalk? (Interview Answer)

AWS Elastic Beanstalk is a Platform as a Service (PaaS) that automatically deploys, manages, and scales the applications without needing to manage the underlying infrastructure.

✅ Why / When Do You Use Elastic Beanstalk? (Interview Answer)

use Elastic Beanstalk when:

✔ 1. We want to deploy applications quickly without managing servers

It automates provisioning of EC2, ELB, Auto Scaling, and monitoring.

✔ 2. we focus is on writing code, not infrastructure

Perfect for developers who don't want to handle DevOps tasks.

✔ 3. we need automatic scaling & monitoring

Top 5 features / advantages of AWS Elastic Beanstalk

✅ 1. Fully Managed Deployment

✅ 2. Automatic Scaling

✅ 3. Built-in Monitoring & Health Checks

✅ 4. Easy to Use & Quick Setup

✅ 5. No Additional Cost

✅ AWS Lambda (Interview Answer)

AWS Lambda is a serverless compute service that lets you run the code without provisioning or managing servers.
You upload your function, and Lambda runs it automatically when triggered.

- No servers to manage
- Automatically scales based on requests
- Pay-per-use (milliseconds billing)

✅ Example 1: Resize Image When Uploaded to S3

When a user uploads an image to an S3 bucket → Lambda automatically resizes it and stores a smaller version.

✅ What is Serverless in AWS? (Interview Answer)

Serverless in AWS means run applications without managing servers. AWS handles all infrastructure tasks like provisioning, scaling, patching, and maintenance. You only focus on writing code.

Common AWS serverless services:

- **AWS Lambda**
- **API Gateway**
- **DynamoDB**

❓ Is Serverless Truly Serverless? (Interview Answer)

No, serverless is not truly "serverless." Servers still exist, but AWS manages them completely behind the scenes.

You never see or manage:

- EC2 instances
- OS patching
- Scaling

✅ What is Amazon RDS?

amazon RDS (Relational Database Service) is a fully managed database service that automates database setup, backups, patching, monitoring, scaling, and maintenance for relational databases.

It supports popular engines like:

- MySQL
- PostgreSQL

How it is different?

Amazon RDS is a fully managed relational database service, unlike a regular database where you manage installation, backups, scaling.

✅ What is AWS VPC? (Interview Answer)

Amazon VPC (Virtual Private Cloud) is a logically isolated network in the AWS cloud where we can launch the resources like EC2, RDS, and Lambda securely.

✅ Why Do We Need a VPC? (Interview Answer)

We need a VPC to:

✔ 1. Control Network Security

You can control inbound & outbound traffic using **security groups** and **NACLs**.

✔ 2. Isolate the Resources

Each VPC is private—no one can access it unless you allow them.

✔ 3. Customize Your Network

You decide IP ranges, public/private subnets, routing, NAT gateways, etc.

✔ 4. Connect On-Premise to Cloud

Using VPN or Direct Connect, VPC allows hybrid cloud setups.

✅ What is ELB?

ELB (Elastic Load Balancer) is an AWS service that automatically distributes incoming traffic across multiple targets like EC2 instances, containers, and IP addresses to ensure high availability and reliability.

Types of ELB

1️⃣Application Load Balancer (ALB)

2️⃣Network Load Balancer (NLB)

3️⃣Gateway Load Balancer (GLB/GWLB)

4️⃣Classic Load Balancer (CLB) *(Legacy)*

✅ What is Amazon Route 53?

Amazon Route 53 is a highly available and scalable DNS (Domain Name System) service by AWS that translates human-readable domain names (like example.com) into IP addresses.
It also supports domain registration and global traffic routing.

✅ How Does Route 53 Manage Global Traffic?

Route 53 manages global traffic using **routing policies** that direct users to the best possible AWS resource based on **location, latency, health**, or **custom rules**.

✅ AWS IAM (Interview Answer)

AWS IAM (Identity and Access Management) is a service that helps you securely manage access to AWS services and resources. It allows you to control *who* can access *what* in your AWS account.

✅ Key Features of IAM (Interview Points)

- **User Authentication** → Create users with login or programmatic access
- **Authorization** → Grant permissions using policies
- **IAM Roles** → Temporary access for services or other accounts

✅ CI/CD Pipeline (Interview Answer)

A CI/CD pipeline is an automated process that helps developers build, test, and deploy code faster and more reliably.
It stands for:

- **CI → Continuous Integration(code build test)**
- **CD → Continuous Delivery / Continuous Deployment**

The pipeline ensures every code change is automatically validated and deployed with minimal manual effort.

✅ What Happens in a CI/CD Pipeline?

1️⃣Code Commit

Developer pushes code to GitHub/GitLab/Bitbucket.

2️⃣ Continuous Integration (CI)

- Automatically builds the code
- Runs tests

3️⃣ Continuous Delivery (CD)

- Prepares the code for deployment

4️⃣ Continuous Deployment

- Automatically deploys the latest tested code to production
- No manual steps required

✅ Amazon CloudWatch (Interview Answer)

Amazon CloudWatch is a monitoring and observability service that collects and tracks metrics, logs, and events from AWS resources and applications.
It helps you monitor performance, detect issues, and set automated alerts.

- Metrics Monitoring : Tracks CPU, memory (via agent), disk I/O, network, etc.
- Log Management : Collects logs from applications, Lambda, EC2, APIs, etc.
- Alarms : Set alarms to get notified or auto-scale resources.

✅ AWS Storage (Interview Answer)

AWS Storage refers to the collection of cloud storage services offered by Amazon Web Services that allow you to store, manage, and back up data reliably and at scale.

1️⃣ Amazon S3 – Object Storage

Used for storing files, images, videos, backups, logs.

2️⃣ Amazon EBS – Block Storage

Used as a virtual hard drive for EC2 instances.

3️⃣ Amazon EFS – File Storage

Shared, scalable network file system for multiple EC2 instances.

✅ Amazon S3 (Interview Answer)

Amazon S3 (Simple Storage Service) is an object storage service used to store and retrieve any amount of data from anywhere on the internet.

✅ What is AWS EBS? (Interview Answer)

AWS EBS (Elastic Block Store) is a block storage service that provides persistent storage volumes for EC2 instances.

✅ **How Does EBS Support EC2? (Interview Answer)**

1️⃣Acts as the EC2 Server's Hard Drive

2️⃣Provides Persistent Storage 3️⃣Supports Backups via Snapshots

4️⃣Offers High Performance & Scalable Storage

✅ What is Block Storage? (Interview Answer)

Block storage stores data in fixed-sized blocks, similar to a traditional hard drive.
You attach it to a server (like EC2), and it behaves like a disk where you can install OS, databases, and file systems.

✔ Examples

- **AWS EBS**

✅ What is Object Storage? (Interview Answer)

Object storage stores data as objects (file + metadata + unique ID) inside buckets.
It is ideal for storing large amounts of unstructured data.

✔ Examples

- **Amazon S3**

✅ What is a Security Group? (Interview Answer)

A Security Group (SG) is a virtual firewall for EC2 instances that controls inbound and outbound traffic at the *instance level*.
It works on **allow rules only** (no deny rules).

✅ What is a Network ACL (NACL)? (Interview Answer)

A Network ACL is a subnet-level firewall that controls inbound and outbound traffic for the entire subnet.
It supports both **allow and deny rules**.

✅ Amazon SQS (Simple Queue Service)

Amazon SQS is a fully managed message queueing service that decouples applications and stores messages until they are processed.

✔ Key Points

- Pull-based messaging (consumers pull messages)
- Used to **buffer and distribute workloads**

✅ Amazon SNS (Simple Notification Service)

Amazon SNS is a fully managed publish/subscribe (pub/sub) messaging service that sends notifications to many subscribers at once.

✔ Key Points

- Push-based messaging
- Sends messages to multiple subscribers simultaneously

DIfference : SQS is a message queue used for decoupling and asynchronous processing, while SNS is a pub/sub service used for broadcasting notifications to multiple subscribers.

✅ Elastic IP (Interview Answer)

An Elastic IP (EIP) is a static, public IPv4 address in AWS that you can attach to an EC2 instance or network interface to ensure the instance always has the same public IP, even after stopping or restarting it.

✅ Amazon DynamoDB (Interview Answer)

Amazon DynamoDB is a fully managed NoSQL database service that provides fast, predictable performance with automatic scaling. It stores data in key-value and document format and is designed for high availability and low-latency applications.

✅ How do you optimize cost for high-traffic applications on AWS?

Use auto-scaling, right-sizing, Spot/Reserved instances, CloudFront, serverless, and monitoring to reduce cost for high-traffic apps.

✅ AWS Direct Connect (Interview Answer)

AWS Direct Connect is a dedicated, private network connection between on-premise data center and AWS.

AZURE INTERVIEW QUESTIONS

✅ What is Azure?

Azure is Microsoft's cloud computing platform that provides on-demand services like compute, storage, databases, networking, AI, DevOps, and security over the internet on a pay-as-you-go model.

✅ “Your client wants to use their existing on-prem servers with Azure. Which cloud model fits best?”

✅ Best Answer: Hybrid Cloud Model

The cloud model that fits best is the *Hybrid Cloud* model.

Because the client wants to **keep using their existing on-premises servers** *and* also use Azure services, hybrid cloud provides seamless integration between on-prem infrastructure and cloud resources.

✅ What is a Resource in Azure?

A resource in Azure is an individual component that you create and manage, such as a VM, database, storage account, or virtual network.
It is the actual *instance* you deploy inside Azure.

✔ Examples of Azure Resources

- Virtual Machine (VM)
- Storage Account

✅ What is a Service in Azure?

A service in Azure is a broader category or product offering provided by Azure that delivers specific functionality.
A service lets you *create resources* under it.

✔ Examples of Azure Services

- Azure Compute
- Azure Storage

✅ What is a Resource Group in Azure? (Interview Answer)

A Resource Group in Azure is a logical container that holds related Azure resources such as VMs, databases, storage accounts, networks, etc.

✅ Why Do We Use Resource Groups in Projects?

1️⃣ Better Organization of Resources

2️⃣ Simplified Management

3️⃣ Role-Based Access Control (RBAC)

4️⃣ Easy Deployment & Automation

✅ What is an Azure Region?

An Azure Region is a geographic location where Microsoft has multiple data centers.
It is the area where your Azure resources are deployed.

✔ Examples

- **East US**

✅ What is an Azure Availability Zone?

Availability Zones are physically separate data centers within the same Azure Region.
Each zone has its own independent power, cooling, networking.

✔ Purpose

- High availability

✅ What Happens When a Resource Group Is Deleted?

When a Resource Group is deleted in Azure, ALL the resources inside it are deleted automatically.
This includes VMs, storage accounts, databases, VNets, NICs, public IPs — everything.

✔ 1. All resources get deleted permanently.

✔ 2. Resources with soft-delete stay recoverable

✔ 3. Deletion may take several minutes

✅ Q. You want to allow a user to manage resources in one project but not another. What's the best way?

⭑ Best Answer: Use Azure Role-Based Access Control (RBAC) at the Resource Group level.

Assign the user permissions (like Contributor role) ONLY to the Resource Group of the project they should manage.

✅ What are Azure Virtual Machines?

Azure Virtual Machines are scalable, on-demand compute resources in the cloud that function like traditional servers.
They allow you to run Windows or Linux operating systems and install any software you need.

They give you full control over:

- OS
- CPU, RAM
- Storage

✅ When Would You Use Azure VMs in a Project?

1️⃣ When you need full control over the operating system

3️⃣ Hosting web servers, application servers, or backend services

4️⃣ For databases that you want to self-manage

✅ What are VM Scale Sets?

Azure VM Scale Sets (VMSS) are a service that automatically deploys, manages, and scales a group of identical virtual machines.
They allow your application to handle variable traffic by **automatically increasing or decreasing** the number of VMs.

✅ When Will You Use VM Scale Sets in a Project?

1️⃣ When your application needs to scale automatically

2️⃣ When your app must be highly available

3️⃣ Running stateless or identical workloads

✅ What is Azure App Service?

Azure App Service is a fully managed Platform-as-a-Service (PaaS) that lets you build, deploy, and scale web apps, REST APIs, and mobile backends without managing servers.

It supports multiple languages like:

- .NET
- Java

✅ For What Purpose Can You Use Azure App Service?

1️⃣ Hosting Web Applications

2️⃣ Running REST APIs / Backend Services

3️⃣ Hosting Mobile App Backends

✅ What is the difference between Azure VMs and Azure App Services?

⚫ Azure VMs (Virtual Machines)

Feature	Azure VMs	Azure App Service
Server	You manage everything	Fully managed by Azure

Management		
Scaling	Manual / VM Scale Sets	Auto-scaling built-in
OS Access	Full access	No OS access
Deployment	Manual or scripted	Automated (CI/CD)
Best For	Legacy apps, custom software	Web apps, APIs, modern apps
Flexibility	Very high	Limited to supported runtimes
Cost	Higher maintenance	Lower maintenance

✅ What Are Deployment Slots?

Deployment Slots are live environments within an App Service (like “Staging”, “Testing”, “QA”) where you can deploy and test your app before releasing it to production.

Your main app runs in the **Production** slot, and you can have additional slots like:

- Staging
- Development

✅ Why Are Deployment Slots Useful?

1️⃣Zero-Downtime Deployment (Swap Feature)

2️⃣Test in a Real Environment

3️⃣Easy Rollback

✅ Q. Your company wants to host a web app with minimum infrastructure management. What do you choose?

⭐ Best Answer: Azure App Service

✅ Why Azure App Service?

1️⃣Fully Managed Platform (PaaS)

- No server setup
- No OS updates
- No patching

2️⃣ Easy Deployment

3️⃣ Autoscaling + High Availability

✅ What Are Azure Functions?

Azure Functions is a serverless compute service that runs your code automatically in response to events—without you managing servers.
You pay only for the execution time.

It supports triggers like:

- HTTP requests

✅ Why Did You Use Azure Functions?

1️⃣ To Avoid Server Management

3️⃣ Event-Driven Workloads

2️⃣ Cost Efficient

✅ Example 1: File Processing from Blob Storage

Use Case: When a user uploads a file to Blob Storage
Trigger: Blob Trigger
Azure Function Action:

- Validate the file
- Compress/resize it
- Move it to another Blob container

✅ Azure Logic Apps

Azure Logic Apps is a fully managed integration service that helps you automate workflows and connect applications, data, and systems using a visual designer—without writing code.

Example :

"When an email arrives in Outlook with an attachment, a Logic App automatically saves the file to Blob Storage, extracts data, and notifies the team on Teams."

✅ Azure Storage (Interview Answer)

Azure Storage is Microsoft's cloud storage solution that provides scalable, durable, and secure storage for all types of data—files, blobs, queues, tables, and disks.

✅ Types of Azure Storage

1️⃣ Blob Storage (Object Storage)

- Stores unstructured data (images, videos, backups, logs).

2️⃣ File Storage (Azure Files)

- Fully managed shared file system using SMB & NFS.

3️⃣ Queue Storage

- Message queueing for asynchronous communication.

4️⃣ Table Storage

- NoSQL key-value storage.

✅ What is Azure Blob Storage?

Azure Blob Storage is Microsoft's object storage solution for storing large amounts of unstructured data such as images, videos, logs, backups, documents, and big data.

It stores data as **blobs (objects)** inside **containers**, similar to folders.

✅ Types of Blobs

Azure Blob Storage supports three types:

1. Block Blobs

- For text and binary data
- Used for files, documents, media
- Most common blob type

2. Append Blobs

- Optimized for append operations
- Ideal for logging (e.g., app logs)

3. Page Blobs

- Used by Azure Virtual Machine **disks**
- Supports random read/write operations

✅ What is the difference between Azure SQL Managed Instance and Azure SQL Database?

● Azure SQL Managed Instance

A fully managed SQL Server instance in the cloud with almost 100% compatibility with on-prem SQL Server.

✔ Key Points

- Supports **SQL Agent, Linked Servers, CLR**, cross-database queries

- Allows **full instance-level features**

🟢 Azure SQL Database

A fully managed single database or elastic pool designed for cloud-native applications.

✔ Key Points

- PaaS database
- No SQL Agent or instance-level features

Feature	Managed Instance	SQL Database
Compatibility	Almost full SQL Server	Limited (DB-level features only)
Architecture	Instance-level	Database-level
Use Case	Lift-and-shift, legacy apps	Cloud-native, new apps
Network	Deployed in VNet	Public endpoint (can restrict)
SQL Agent	✔ Supported	✖ Not supported
Cross DB Query	✔ Yes	✔ Limited via elastic query
Cost	Higher	Lower

⚫ What is a Virtual Network (VNet)?

A Virtual Network (VNet) in Azure is a private, isolated network in the cloud where you can securely run your Azure resources like VMs, databases, and services.
It is similar to an on-premise network but hosted in Azure.

🟢 Why Do We Need a VNet?

1️⃣ Network Isolation and Security

2️⃣ Connect Resources Securely

3️⃣ Hybrid Connectivity

⚫ What is Azure CDN? (Interview Answer)

Azure CDN (Content Delivery Network) is a globally distributed network of servers that delivers web content (images, videos, CSS, JS, documents) to users with low latency by caching content at edge locations around the world.

When & Why to Use Azure CDN? (Interview Points)

1️⃣ Faster Content Delivery (Low Latency)

2️⃣ Reduce Load on Your Web/App Servers

3️⃣ Improve Global Performance

5️⃣ Secure Content Delivery

What is Azure Key Vault?

Azure Key Vault is a cloud service used to securely store and manage secrets, keys, passwords, certificates, API keys, and connection strings.
It prevents sensitive information from being stored in application code or config files.

Why Do We Use Key Vault?

- Centralized storage for secrets
- Secure access control using Azure AD
- Automatic rotation of keys/secrets

Example: When I Used Key Vault in My Project

"In my project, we stored database connection strings, API keys, and storage account keys inside Azure Key Vault instead of keeping them in config files. Our App Service accessed the secrets securely using a Managed Identity, so no credentials were hard-coded or exposed."